

Zarif Fida Chowdhury

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SUMMARY

Software engineer (B.S. Computer Science, Miami University, expected September 2026) building full-stack systems with the security work done up front: server-side key handling, session authentication, validated input. Delivered Capstone GPT, a course-grounded RAG assistant, solo. Open to new-grad roles in software, security, and AI.

PROJECTS

Capstone GPT Miami University · github.com/zariffidachowdhury/capstone-gpt Spring 2026

- Course-grounded RAG assistant for Senior Design students (PHP 8, MySQL, JavaScript, Dify): answers from the syllabus and course materials, not the open internet. Delivered solo, via GitLab issues, branches, and merge requests.
- LLM API key kept server-side behind one PHP control point that handles validation, prompt composition, and logging.
- Session authentication with bcrypt (password_hash) and 256-bit tokens from random_bytes(32). Input validated server-side before it reaches the retrieval workflow.
- Ten REST routes across seven PHP endpoints over a normalized five-table MySQL schema: conversation logs, feedback capture, instructor analytics, CSV export.

secure-rag-guardrails open source · github.com/zariffidachowdhury/secure-rag-guardrails Jun 2026

- Input gate, retrieved-context fence, and output redaction for LLM apps, mapped to the OWASP Top 10 for LLMs. One file of standard-library Python, 13 tests in CI, every decision returns the rules that fired.

zariffidachowdhury.github.io personal site · source on GitHub Sep 2026

- Hand-written HTML, CSS and JavaScript under a strict Content Security Policy. Every file's SHA-256 is pinned in the page, enforced through Subresource Integrity and re-checked in the visitor's browser.
- In-browser Feistel network visualizer and ray tracer in plain JavaScript. Project claims unfold to the exact source lines, pulled from the repos by a build script.

EXPERIENCE

CSE Student Ambassador Miami University, Oxford, OH Oct 2025 - present

- The department's face at open houses of 200+ guests each semester: event logistics, breakout sessions and Q&A panels on the curriculum, and course-selection mentoring for admitted students.

Cybersecurity Intern SMART Technologies Ltd., Dhaka, Bangladesh Jun - Aug 2024

- Researched SIEM systems and presented findings to the security-monitoring team. Enterprise networking: port forwarding and device configuration.
- Completed Cisco Linux Essentials: Linux administration and command-line work.

Software Developer Intern SMART Technologies Ltd., Dhaka, Bangladesh Jun - Aug 2023

- Built modular Python backend prototypes for internal data-processing tools. Designed PostgreSQL schemas and indexing that improved query performance by 15%.
- Worked in an Agile team: daily stand-ups, code review, presenting solutions to stakeholders.

HANDS-ON COURSEWORK

CSE 386	Computer Graphics. C++ ray tracer: Phong lighting, triangle, cylinder and disk intersections, texture mapping, Z-buffer hidden surfaces, alpha blending, spotlight cones, GLM transformation pipelines.
CYB 236	Data Security. Designed and implemented ZFC-Cipher, a 64-bit Feistel block cipher. Steganography extraction, a Saltzer & Schroeder analysis, and a physical security assessment of a campus building.
CYB 334	Network Security. pfSense firewall (LAN/WAN/DMZ rules, split DNS), IPsec/IKEv2 VPN, Snort, Kiwi Syslog and Splunk monitoring pipeline. Metasploitable 2 exploitation, SSH log forensics, AES decryption.
CYB 331	Software Security. Threat modeling, OWASP Top 10 (CSRF, XSS, injection), SAST/DAST and fuzzing, secure code review, software supply-chain security.

EDUCATION

Miami University B.S. Computer Science, Oxford, OH Expected September 2026

Additional coursework: Computer Networks, Database Systems, Systems I & II, Algorithms, Machine Learning.

Cybersecurity Club member since 2022. Cisco Linux Essentials (2024). CompTIA Security+ and AWS Solutions Architect Associate in progress.

SKILLS

Languages	Python, Java, C++, C, PHP, JavaScript, SQL
Security	OWASP Top 10, threat modeling, Burp Suite, Wireshark, tcpdump, Nmap, Metasploit, OpenSSL, Kali Linux, pfSense, Snort, Splunk
Systems & AI	Linux and Bash, Docker, AWS, Git, GitLab, MySQL, PostgreSQL, REST, RAG, Dify, LLM APIs